SHUBHAM GOEL

CSE Undergraduate @ IIIT Hyderabad

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EXPERIENCE

Undergraduate Researcher

Center for Visual Information Technology (CVIT)

May 2025 - Present Hyderabad

Advisor: Prof. C. V. Jawahar

- Working on **AI-Generated Image Detection and Attribution**, investigating techniques to distinguish AI-generated images from non-AI-generated images and identify their generative sources.

System SWE Intern

NVIDIA Graphics Pvt. Ltd.

May 2026 - July 2026 Hyderabad

- NvGPU QNX driver refactoring. Security tests automation in GVS.
- Received return offer .

PUBLICATIONS

- **How Does India Cook Biryani?**
  Shubham Goel, Farzana S., C. V. Rishi, Aditya Arun, C. V. Jawahar. *ICVGIP 2025*. Developed a large-scale dataset of 120 biryani preparation videos across 12 regional styles, a VLM-based framework for fine-grained procedural comparison, and a multi-level QA benchmark for evaluating multimodal procedural reasoning.

PROJECTS

Distributed Event Log | *C++20, distributed systems*

- Built a Kafka-inspired partitioned, replicated event log in C++20 with consumer groups, dynamic rebalancing, and epoch-based fencing of stale consumers.

KRAKEN: Knowledge-Graph VQA | *PyTorch, Transformers*

- Built a multimodal QA model that fuses a GAT graph encoder, a frozen ViT with a Q-Former, and the question text, decoded by BART.
- Extended WebNLG into 35K+ rendered graph images with auto-generated QA pairs.

Network File System | *operating systems and networks*

- Built a distributed NFS with naming server, storage servers, and client components, integrating concurrent file operations, LRU caching, data replication, and audio streaming.
- Optimized file path resolution using tries and hashmaps.

Enhancements in xv6 | *operating system and networks*

- Implemented custom system calls, UNIX-like signal handling, and advanced scheduling algorithms (MLFQ, Lottery Scheduling).
- Integrated Copy-on-Write (CoW) fork for optimized memory usage.

Raytracer | *C++, Computer Graphics*

- Implemented a recursive ray tracer in C++, supporting ray-object intersections, surface normals, diffuse (Lambertian), metallic, and dielectric materials with reflections and refractions.

EDUCATION

Btech. CSE (Honours)

IIIT HYDERABAD

August 2023 – Present

CGPA: 8.76/10

SKILLS

C C++ Python Javascript HTML CSS MongoDb MySQL TailWindCss React NextJs ExpressJs Flask Pandas Numpy OpenCV Pytorch Git Github Postman

KEY COURSES TAKEN

Discrete Maths, Linear Algebra, Probability & Statistics, Data Structures and Algorithms, Operating Systems & Networks, Software Engineering, Distributed Systems, Statistical Methods in AI, Computer Vision , Advanced Natural language processing, Data and Applications, Computer Graphics

POSITIONS OF RESPONSIBILITY

- **Teaching Assistant**, IIIT Hyderabad
  - Operating Systems and Networks — Monsoon 2025
  - Introduction to Software Systems — Spring 2026
  - Operating Systems and Networks — Monsoon 2026

ACHIEVEMENTS

- Secured AIR 825 in JEE Mains
- Secured AIR 3482 in JEE Advanced
- Deans List Award , IIITH
- Research Award , IIITH

EXTRACURRICULARS

- District-level Football Player
- Taekwondo Black Belt – Represented in multiple State and National level tournaments